

Arnav Verma

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EDUCATION

Vellore Institute of Technology, Vellore
B.Tech in Computer Science and Engineering

Expected 2026
CGPA: 8.85 / 10.0

EXPERIENCE

DexiChat

Jan 2026

Fullstack Developer (Freelance)

Remote

- Built and maintained **31 GraphQL operations** (queries/mutations) across Shopify Admin + Storefront APIs spanning products, variants, customers, carts, checkout, and orders on a **Node.js/Express** multi-tenant SaaS backend.
- Improved LLM commerce flow with **11 Shopify tool definitions** (search, cart, order prep/confirm, availability) across **4 Shopify service modules** using tool-driven **LLM orchestration**.
- Enhanced product search with **hybrid retrieval** — Storefront predictive search + semantic vector fallback (**pgvector**) — improving handling of plurals, typos, and natural-language queries.

DexoDash

May 2025 – July 2025

Fullstack Developer (Internship)

Remote

- Built a scalable Next.js 15 app with **55+ API endpoints** and **23+ PostgreSQL models**, including cart and checkout flows.
- Integrated Google Gemini for AI-powered content generation across **46+ admin components** with live rich text editor functionality.
- Implemented full **i18n** with English/Hebrew support, RTL layout, currency conversion, and **RBAC with fine-grained permissions**.

PROJECTS

StudyLens (AI Study Monitor)

Technologies: Python, OpenCV, dlib, DeepFace, Tkinter, FastAPI, PyInstaller

Live Demo | GitHub

- Built desktop app analyzing student focus via real-time local webcam processing (privacy-focused).
- Integrated real-time drowsiness detection (Eye Aspect Ratio using dlib) and facial emotion analysis (DeepFace).
- Implemented optional session statistics upload to a secure FastAPI backend; packaged into distributable macOS executables via PyInstaller.

Real-Time Drowsiness Detection System (Collaborative)

Docker, AWS EC2, Flask, Socket.IO, Bash, Python, systemd

GitHub

- Developed a containerized computer vision application using Docker and Docker Compose, implementing multi-stage builds, health checks, and volume mounts for persistent model storage.
- Automated AWS EC2 deployment using Bash scripts, configured systemd services for auto-restart, and applied UFW firewall rules to enhance production security.
- Architected a multi-device monitoring system with a centralized Flask dashboard, real-time streaming via Socket.IO, and RESTful APIs for reliable device communication.

TECHNICAL SKILLS

Languages: Python, C++, Java

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, Flask, FastAPI

Databases: MySQL, PostgreSQL, pgvector

Libraries: NumPy, Pandas, Matplotlib, OpenCV

Tools & Platforms: Git, GitHub, Docker, Vercel, AWS, GraphQL

ACHIEVEMENTS

Patent Application Published

Jan 2026

System and Method for Lightweight Machine Learning Based Intrusion Detection in Agricultural IoT Networks

Indian Patent Application No. 202541131653 A — Filed Dec 2025, Published Jan 2026

CERTIFICATIONS

DevOps Fundamentals

Credential ID: CEDEV2IN

Issued by IBM

DevOps, Agile and Design Thinking

Credential ID: CEDEA1IN

Issued by IBM